

Assignment.

- 1) What is the output of the following, and why?

```
S = "python"
```

```
S[0] = 'p'
```

```
S[-1] = 'n'
```

```
Print (S[0], S[1], S[6]) S[6] -> IndexError  
                        (out of range)
```

- 2) What is the difference between positive and negative indexing? Give example.

positive indexing starts from left (0, 1, 2...).

Negative indexing starts from Right (-1, -2...).

example: S = "Hello"

```
S[-2] = 'l'
```

- 3) What happens when index is out of range?

indexing -> gives IndexError

slicing -> returns available part without errors.

```
S = "code"
```

```
S[10] -> Error
```

```
S[1:10] = "ode"
```

4) Find first and last occurrence of 's' in "mississippi".

First 's' $\rightarrow$ index 2

Last 's' $\rightarrow$ index 6.

5) What is the output?

S = "abcdef"

print (s[-1], s[-3], s[-6])

f d a

6) Define slicing syntax ..

Syntax : s[start : stop : step]

* start $\rightarrow$ beginning index

* stop $\rightarrow$ ending index (excluded)

* step $\rightarrow$ increment value

7) Find outputs:

`S = "Helloworld"`

`S[0:5] -> "Hello"`

`S[5:] -> "World"`

`S[:5] -> "Hello"`

`S[:] -> "Helloworld"`

8) What is the output?

`S = "python"`

`print(S[4:7])`

Empty string ""

9) How to get last 3 characters of a string?

`S[-3:]`

10) Find outputs:

$s = \text{"abcdefgh"}$

$s[::1] \rightarrow \text{"abcdefgh"}$

$s[::2] \rightarrow \text{"aceg"}$

$s[1::2] \rightarrow \text{"bdfh"}$

(11) How to reverse a string?

$s[::-1]$

(12) Find outputs:

$s = \text{"abcdefgh"}$

$s[::-1] \rightarrow \text{"hgfedcba"}$

$s[::-2] \rightarrow \text{"hfdab"}$

$s[6:1:-1] \rightarrow \text{"gfedc"}$

(13) Extract only digits from string:

`S = "a1b2c3d4e5"`

`S[1::2] -> "12345"`

(14) What is negative step in slicing?
negative step moves from right to left.
Start index must be greater than stop index.

Example : `S[5:1:-1]`

(15) How do slices handle start and stop values that exceed string boundaries with a negative step?

`S = "hello" print (S[100::-1])`

`print (S[::-100])`

(16) How do you extract the middle portion of a string dynamically, regardless of its length?

S = "abcdefghij" # 10 chars # Extract characters from index 1 2 3 4 5 6 7 8
2 to record last (exclusive)

Soln: [2:9]

(17) How do you use slicing to check if a string is a palindrome in a single line?

S = "racecar"

check palindrome using slicing

Soln: [S[::-1]]

palindrome == True.

(18) What is the output of the following and why does python not raise an error?

```
s = "Hello" ; print (s[1:100]) ;  
print (s[-100:3]) ; print (s[100:200])
```

soln:

```
print (s[1:100]) = ello  
print (s[-100:3]) =  
print (s[100:200]) = " "
```

Because it take only exist index value so it is not raise an error.

(19) How do you replace a portion of a string using slicing (since strings are immutable)?

```
s = "Hello world"  
# Replace "world" with "python" using  
slicing and concatenation.
```

soln:

```
print ('HelloWorld + s')
```

```
s = HelloWorld.
```

```
# word = python ; # s = python.
```

```
print ('Hello + s')
```

```
Hellopython.
```

(20) How does python's slice () object work, and how can you use it as a reusable slices?

```
s = "abcdefghij"
```

```
0123456789.
```

```
my - slice = slice (1, 8, 2).
```

```
print (s [my - slice])
```

What is the output, and how can this be reused on

other strings?

soln:

```
s [b, i, c].
```